

ARM PrimeCell

External Bus Interface (PL220)

Release Note

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General suggestion for additions and improvements are also welcome.

Feedback on the ARM PrimeCell External Bus Interface

If you have any comments or suggestions about this product, please contact your supplier giving:

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Change history

Issue	Date	Change
A	18 11 2002	First release.

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1 PRODUCT DELIVERABLES

1.1 ARM Part Numbers for this product

Table 1.1-1 lists the ARM part numbers for the individual deliverables that form part of the delivery for this ARM PrimeCell peripheral:

Product deliverables for the ARM PrimeCell PL220 peripheral	Document number	Product code
Synthesizable VHDL RTL	N/A	PL220-MN-23001-R0p0-00ltd0
Synthesizable Verilog RTL	N/A	PL220-MN-22001-R0p0-00ltd0
Synthesis Scripts (Synopsys Design Compiler, Avant! CB18 cell library)	N/A	PL220-MN-01002-R0p0-00ltd0
Testbench - Functional Verification, VHDL	N/A	PL220-MN-23010-R0p0-00ltd0
Testbench - Functional Verification, Verilog	N/A	PL220-MN-22010-R0p0-00ltd0
Test vectors-BusTalk, functional verification	N/A	PL220-VE-01010-R0p0-00ltd0
ARM PrimeCell External Bus Interface (PL220) Technical Reference Manual (PDF)	ARM DDI 0249 A	PL220-DA-03001-R0p0-00ltd0
ARM PrimeCell External Bus Interface (PL220) Design Manual (PDF)	PL220 DDES 0000 A	PL220-DC-02002-R0p0-00ltd0
ARM PrimeCell External Bus Interface (PL220) Integration Manual (PDF)	PL220 INTM 0000 A	PL220-DC-10002-R0p0-00ltd0
ARM PrimeCell External Bus Interface (PL220) Release Note (PDF)	PL220 RELN 0000 A	PL220-DC-06002-R0p0-00ltd0
ARM PrimeCell External Bus Interface (PL220) Errata Notice (PDF)	PL220 DEI 0000 A	PL220-DC-11001-R0p0-00ltd0
ARM PrimeCell External Bus Interface (PL220) Userguide (PDF)	PL220 USGD 0000 A	PL220-DC-08001-R0p0-00ltd0

Table 1.1-1: ARM part numbers for ARM PrimeCell EBI PL220

Note Only the required deliverables will be supplied, as per the contractual agreement.

1.2 Reference Cell Library

A reference cell library (Avant! CB18 0.18µm standard cells) is supplied with this peripheral.

PL000-BU-00005-0.00/Avanti/CB18_2000.3.tgz
 PL000-BU-00005-0.00/Avanti/CB18_2000.3.lst

2 INSTALLATION

2.1 Introduction

Intellectual Property (IP) deliverables are collectively delivered as a single UNIX zipped tar file which has been security encrypted.

An ARM PrimeCell peripheral may be one of several products delivered as part of this single encrypted file. Each ARM PrimeCell peripheral comprises a set of zipped tar files which correspond to a unique part number quoted in the contractual agreement document. A corresponding list file details the files in each tar file

These installation instructions cover the following platform/operating system:

- UNIX

2.2 Disk space required

This installation will require 1.5 Mbytes of free disk space.

2.3 Installation procedure

You will have a single tarred, zipped and encrypted file of following format:

E.g.

Company Name-PrimeCell-PL220–Bundled-arm-download-WWWW.tgz.des

The installation procedure is summarised below:

2.3.1 Unpacking the shipment

The following steps describe how to unpack the separate products delivered in this shipment.

The first three steps may have already been done, ie the shipment from ARM may have already been decrypted and unzipped. If this is the case then go to step 4, else start with step 1.

1. Relocate the shipment file

Relocate the encrypted file (for example <file>.tgz.des, if DES encryption has been used) to an appropriate location that has sufficient free disk space if PGP encrypted will need the relevant keys.

2. Decrypt file

Follow the accompanying instructions for decryption of the file supplied, to generate a decrypted, compressed, tar file <file>.tgz

3. Uncompress file

Execute the UNIX unzip command to obtain the tar file <file>.tar:

```
gunzip <file>.tgz
```

where <file> is the name of the file supplied.

4. Extract tar files

To extract the PrimeCell, execute the UNIX tar extract command:

```
tar -xvf <file>.tar
```

This will generate the following three files

ARM_MANIFEST_220X.TXT – which lists the PrimeCell files delivered

ARM_DELIVERY_220X.TXT – which lists the ARM parts delivered

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And will unpack the PrimeCell into the top-level peripheral directory (e.g. ebi_PL220)

The peripheral development environment is stored in two directory structures, a peripheral specific directory structure and a project global directory structure. The top level directory structures are shown in **Figure 2-1**.

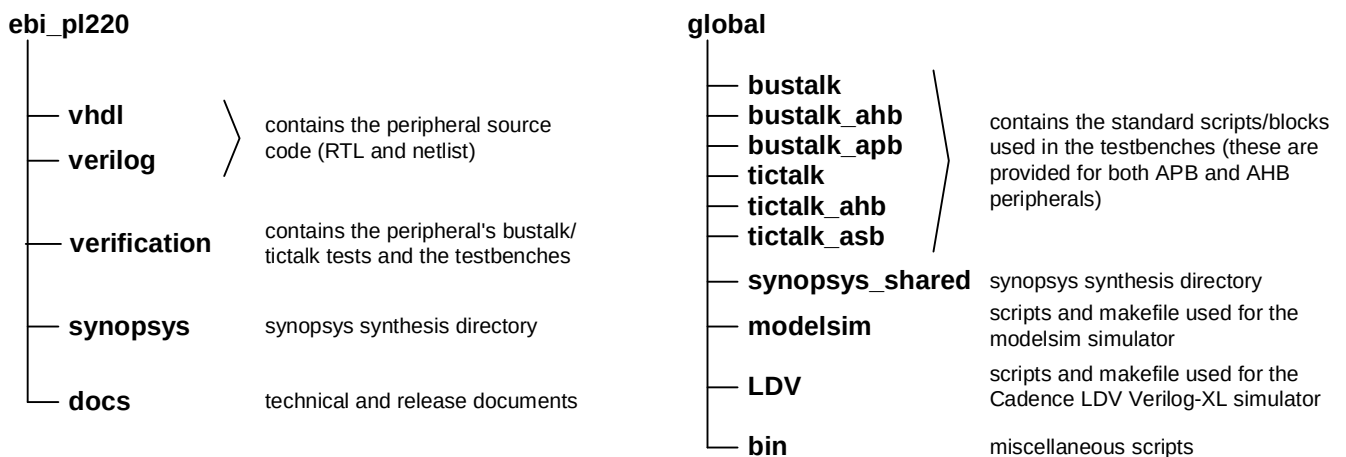


Figure 2-1. Peripheral Development Environment Directory Structure

2.3.2 Installation of the reference cell library

The reference cell library should be copied manually and tar extracted in the directory containing the PrimeCell peripherals. An example is shown below to copy the Avant! CB18 2000.3 cell library to the directory /ARM/PrimeCell_peripherals

```
cd /ARM/PrimeCell_peripherals
cp PL000-BU-00005-0.00/Avanti/CB18_2000.3.tgz .
```

Execute the UNIX unzip and tar extract commands as below:

```
gunzip CB18_2000.3.tar.gz  
tar -xvf CB18_2000.3.tar
```



3 DOCUMENTATION

3.1 Adobe PDF

Documents are supplied as "Adobe PDF" (Portable Document Format) files. These files are readable on most common computer platforms and operating systems using the appropriate file reader. A suitable file reader can be downloaded from the Adobe World Wide Web site at <http://www.adobe.com>. Select "Acrobat Reader" and download the reader for your computer platform/operating system.

3.2 Peripheral Documents Supplied

The following documents are supplied for the ARM PrimeCell External Bus Interface PL220:
The documents files are located in the ebi_PL220/docs/ directory.

ARM PrimeCell External Bus Interface PL220 Technical Reference Manual ARM DDI 0249

The technical reference manual provides details for programming and interfacing to this ARM PrimeCell peripheral block.

ARM PrimeCell External Bus Interface PL220 Integration Manual PL220 INTM 0000

The integration manual provides information required to integrate this ARM PrimeCell block within a typical industry standard ASIC design flow.

ARM PrimeCell External Bus Interface PL220 Design Manual PL220 DDES 0000

The design manual describes the implementation details of this ARM PrimeCell block and the testbenches and test programs used for functional verification.

ARM PrimeCell External Bus Interface PL220 Errata Notice PL220 DEI 0000

The errata notice describes details of any known issues with this ARM PrimeCell block and any workarounds for these issues. In most cases this document will be blank or may not be present.

ARM PrimeCell External Bus Interface PL220 User Guide PL220 USGD 0000

The userguide describes the instructions on performing simulations and synthesis for this ARM PrimeCell block.



4 TRIAL TESTING

4.1 Product Maturity

This product is newly developed IP and approved for final release.
At the time of release, it has not been implemented in production silicon.

4.2 Technical data summary

The following is a summary of the key technical points from testing of the ARM PrimeCell External Bus Interface PL220.

	VHDL RTL	Verilog RTL
Cell library	Avant! Passport 2000.3 CB18 v1.0 (0.18 μ m)	Avant! Passport 2000.3 CB18 v1.0 (0.18 μ m)
Total cell area, without scan, NAND-2 equivalents	1776	
Net interconnect area, without scan, NAND-2 equivalents	597	
Total cell area, scan inserted,NAND-2 equivalents	2231	2154
Net interconnect area, scan inserted,NAND-2 equivalents	724	642
Clock frequency used	190.5 MHz	190.5 MHz
Scan insertion figure % (number of 'D' flip-flops replaced)	100 % (146)	% ()
Estimated scan fault coverage, collapsed.	100%	
Synthesis and scan test tools and version	Synopsys Design Compiler and Test Compiler 2000-11-sp1	Synopsys Design Compiler and Test Compiler 2000-11-sp1
Simulation tools and versions	Model Technology Inc, ModelSim v5.5d	Model Technology Inc, ModelSim v5.5d Cadence Verilog-XL 3.2
Global Peripheral Development Environment Version	Release REL1v7	

4.3 BusTalk Functional block verification test scenarios

Functional block verification has been performed with the following scenarios using the BusTalk testbench with free-running clocks.

4.3.1 Synchronous clocks

SMCCLK/PCLK	Simulations
190.5 MHz (ns period)	VHDL and Verilog, RTL and gate-level min/max SDF

For further details of functional verification testing, refer to the ARM PrimeCell External Bus Interface (PL220) Design Manual.

5 DIFFERENCES FROM PREVIOUS REVISIONS

First release.

6 KNOWN ISSUES

At first release r0p0-00ltd0 there are no known issue with PL220

The Errata notice will be updated with any issue found subsequent to the release.

7 REVISION HISTORY

Date	Revision	Description
10 Dec. 2002	R0p0-00ltd0	Initial Release 1.0